ERC Consolidator Grant 2027

Research proposal [Part B1]

A New Web: Bridging NFPs for Heterogeneous Societal Requirements

ISLANDS

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Abstract

The World Wide Web scaled to billions of users through architectural decisions that gave up stable references, versioning, and bidirectional links. Its downsides show today, where information has societal consequences: a fabricated audio clip of a party leader circulated, unverifiable, during Slovakia's 2023 pre-election silence; over half the URLs in UK parliamentary answers of 1997–2006 no longer resolved by 2007; courts in Europe and beyond have received over 1,200 filings with AI-fabricated citations; and copyright courts from Munich to New York must reconstruct attribution trails the Web never recorded. The Web's architecture hence deferred, rather than avoided, the cost of coordination, and society now pays it.

ISLANDS extends today's Web with an architecture organized around *non-functional properties* (NFPs). *Islands* are bounded networked information spaces whose members choose the NFPs their context requires and provide them within the island, while islands are kept interoperable and backwards-compatible through today's Web's low-coordination architecture. With affected communities, we first establish which islands society requires, starting from legal citation and copyright, supply-chain carbon accounting, public-interest social media, signed research claims, and fixed industrial installations. We then derive the link classes these islands need, from *immutable, span-addressed versions* to references resolved through *secure multi-party computation*, and research how NFPs can translate at island borders. Our scientific core is a *theory of chosen guarantees* predicting which NFP bundles an island of a given size and threat model can hold, at what coordination cost. We embody it in an open reference architecture and validate it with each island's community. Together, this theory and architecture will give society an extended Web whose guarantees are chosen by the communities that require them, and on which courts, parliaments, and markets can rely.

Cross-panel / interdisciplinary explanation

The proposal is computer-systems research and is submitted to **PE6 (Computer Science and Informatics)**. It spans four PE6 descriptors: PE6_10 (Web and information systems) as its core, PE6_2 (distributed systems), PE6_5 (cryptology, security, privacy; the confidential references and border checks of O3 and O4), and PE6_7 (artificial intelligence, intelligent systems, multi-agent systems; the agent layer of O4). We expect the scientific contribution to lie in the composition, which none of these sub-fields produces by itself.

Two validation domains and one objective also engage legal and governance scholarship (**SH2**, descriptor SH2_4, legal studies): we design the legal-citation island with legal practice and the social-media island against the data-access and transparency duties of the Digital Services Act, and the stakeholder analysis (O1, O5) builds on legal scholarship on architecture as regulation, on RFC 9518's centralization analysis, and on the PI's published work on DSA auditing and transparency regulation. The same analysis covers the business models, costs, and incentives that sustain each island, so that the proposal is interdisciplinary in its assessment of the islands while its methods and deliverables remain in computer science. External SH2 expertise, including on the economics of institutions, would sharpen the assessment of O1 and O5; the methods and deliverables are PE6, and SH2 is flagged as the only secondary panel.

Section a: Part I of the Scientific Proposal

1. The problem and the idea

The World Wide Web's original design favored simplicity, decentralization, and low coordination overhead, which enabled explosive adoption but sacrificed stable references, built-in versioning, and native bidirectional linking. The hypertext community had debated all three as core issues from 1988 to 1992 [1, 2]. This was a deliberate choice: Berners-Lee's 1989 proposal required that the system "allow existing systems to be linked together without requiring any central control or coordination" [3]. When Fielding and Taylor later defined the Representational State Transfer (REST) architectural style, they clarified the trade-offs that the resulting architecture yields: scalable component interactions, uniform interfaces, independently deployable components, and intermediaries that can be inserted anywhere [4]. These priorities suited early document-sharing systems and enabled the Web's scaling up to billions of users and devices within three decades. We argue that the Web's trade-offs are, however, too rigid for today's use cases. Publishers require traceable dereferencing, supply chain management requires aggregation over confidential data, and courts and parliaments require stable, specific citations. Users require sovereignty over their data, and society requires trusted content and accountable authorship.

Over the past twenty years, we have kept the Web's architecture fundamentally static and such properties lacking; however, platforms have responded: Accounts, persistent storage, search, reputation, payment, and a consistent shared state do not exist between independent Web sites. Platforms provide them inside proprietary stores into which they draw the Web's content, users, and interactions [5, 6, 7], and they extend into the rest of the Web through APIs and plug-ins on their own terms [8]. Users then rebuild continuity themselves through reposts, screenshots, and off-platform link hubs [9], yielding unsatisfactory mitigation. Berners-Lee warned in 2010 that social networks were walling off the information their users posted [10], and RFC 9518 describes the general mechanism: functions built on open protocols still centralize through network effects and switching costs [11]. The consequences reach the public sphere: Engagement-optimized ranking amplifies some political content over other content [12]. The EU's Digital Services Act now obliges the largest platforms to assess such systemic risks and to open their data to vetted researchers [13]. This obligation remains hard to operationalize, because researchers must specify data whose existence they cannot observe [14]. The Commission's first fine under the Act, €120 million against X in December 2025, concerned a "verified" badge that anyone could buy without an identity check and terms that barred researchers from public data [15]. The Web itself authenticates domains through TLS certificates, but it offers no native way to state verifiably who stands behind a document or an account, or what qualifies them [16]. The W3C now lists among its Ethical Web Principles that the Web "makes it possible to verify information", since "the public needs verifiable source and context information to recognize trustworthy web publishers and content" [17]. Even the data that platforms do release differs from what their users see [18, 19]. It can also change silently. In August 2026, six months after publication, TikTok repointed the download link of its DSA transparency report to a corrected version without announcement; the correction lowered the reported terrorist-content notices from 77,729 to 43,119 [20]. Press publishers face the same opacity. In October 2026, the Paris Court of Appeal upheld an order that Meta negotiate neighboring-rights payments with French media in good faith; it found that Meta had supplied only fragmentary, opaque, and unverifiable data on its use of their content [21]. Legal scholarship treats such architectural choices as regulation in their own right [22, 23, 24]. Generative AI now repeats the pattern: answer engines and agentic browsers return answers in place of links and so control access to the Web's content [25]. The W3C Technical Architecture Group (TAG), in contrast, holds that a user agent acting on its user's behalf owes "meaningful visibility into the sources it relied on" [26]. Highlighting real-world consequences, Reddit announced in October 2026 a move to a primarily logged-in experience, to protect its data from scraping [27] after AI answers cut its search referrals by more than a million per week. This shows that, today, the properties that high-stakes communities require are available, but inside closed platforms and under these platforms' control – hardly an Open Web.

Society pays for these omissions: In 2007, the UK government found that more than half of the roughly 4,000 URLs cited in ministerial answers to Parliamentary Questions since 1997 no longer resolved, and it built a redirect service into its national Web archive in response [28]. One in five scholarly articles cites Web content that has since rotted [29]. Two days before Slovakia's 2023 election, a fabricated recording of a party leader discussing vote-rigging circulated with no provenance that voters could check [30]. From February 2027, batteries sold in the EU (and, soon after, products from clothes to detergents) must carry a digital passport with carbon-footprint and due-diligence data assembled from suppliers along the chain [31]. Law shows the cost most sharply, because there a reference's stability is part of the claim it supports. The EU has given judgments and legislation persistent identifiers through ELI [32, 33], but these name documents, not cited passages or versions; half the URLs cited in U.S. Supreme Court opinions no longer resolve, and Perma.cc archives cited pages one citation at a time [34]. By early 2026, more than 1,200 court decisions worldwide concerned AI-

fabricated citations [35]. The High Court of England and Wales warned that citing fabricated authorities may lead to referral to the regulator or to contempt proceedings [36]. Content provenance is being moved into camera hardware [37] and into EU law [38, 13], and the Munich Regional Court held in November 2025 that a language model's memorization of song lyrics infringes copyright [39], a similar question to what *The New York Times v. OpenAI* is litigating in the U.S. [40, 41]. *What the cases have in common is that coordination cost that the Web avoided was deferred, not eliminated; society now pays it piecemeal, one incident at a time.*

Why now. Building these properties into the architecture pays off today, where it did not in 1989. First, the coordination they require has become far less costly. Content addressing [42], *verifiable credentials*, a W3C Recommendation since 2025 [16], and practical *secure multi-party computation* [43] are standardized and deployed, and every EU Member State must offer a European Digital Identity Wallet by 24 December 2026 [44]. Second, demand for the same guarantees now arrives from many directions at once. The DSA requires attribution and research access [13], the Data Act portability [45], the AI Act respect for machine-readable reservations against text and data mining [46], and the battery passport and CBAM verifiable supply-chain data [31, 47]. Today, every platform implements each of these obligations separately, and every regulator checks them separately: N platforms and M obligations yield N × M integrations. An architecture specifies each obligation once, as an NFP bundle, and lets each platform implement it once, which turns N × M into N + M. Third, platform-level solutions fail visibly. A platform can withdraw or silently change what it offers, as the cases above show, and machines, which now produce most Web traffic [48], cannot check a platform's promises. Finally, the protocols that will carry these guarantees are being standardized now, in the W3C groups on the social Web, storage, credentials, and agents and in agent protocols such as MCP [49] and x402 [50]. Once they settle, the opportunity to build the guarantees in closes, as it did in the 1990s.

The idea. ISLANDS proposes to *extend* today's Web's architecture and build selected properties into the Web itself. Communities that require these properties can then select their *own property configuration*, i.e., a set of desired *non-functional properties* (NFPs) that apply to their part of the Web. Such communities are not small. One example is the *copyright community*, i.e., all stakeholders with an interest in having content rights preserved by the architecture, from authors and publishers to collecting societies and courts. Another is an industry's supply chain that must account for its carbon emissions. *Islands* are bounded networked information spaces, such as a legal-citation community, a supply chain that accounts for its carbon emissions, or a community of public-interest publishers on social media. Their members share stakes high enough to justify the cost of coordination, and they settle it through an ex-ante selected architectural trade-off rather than ex-post patching or litigation. *The New York Times*, for instance, is litigating the use of its articles by AI models [40]. It could choose to publish its articles into an island whose NFPs bind every quotation to its source through traceable *transclusion* and carry the paper's reservation against text and data mining [51, 46] in each reference. Readers and AI models that accept these terms enter the island. A paywall achieves less: it controls access on one platform, and once an article has been read, its quotations and copies leave without terms or provenance. In the island, the terms and the attribution travel with every quotation to every participating site and agent, and reuse becomes verifiable without any single platform controlling the articles. Each island selects the NFPs its stakes demand and provides them jointly within its scope, while islands interoperate with one another and with the open Web through the Web's existing low-coordination mechanisms that permit planetary scaling. Within an island, references can for instance be span-level, version-pinned, bidirectional, and verifiable, similar to how Xanadu specified them [52, 53]. Across the boundary, they degrade to ordinary Web links, so that every island remains backwards-compatible with today's Web. Islands can also move away from the Web in the opposite direction. An island of fixed, high-speed installations may bind its interfaces once, at design time, in a binary encoding, much as a linker resolves a program's symbols when it builds the program [37, 38]. Entering an island is an explicit admission step: a client, human or agent, is shown the island's NFP set and accepts the obligations of membership, and this is where the island applies its attribution regime. Through ISLANDS, we further explore NFPs that have not yet been proposed on an architectural level. These include *confidentiality-preserving references* that resolve to a value that members can verify was correctly derived from data they may not read; this generalizes *secure multi-party computation* (SMPC), which has run in production since 2008 [54, 55]. They also include *purpose binding* which attaches the terms of a disclosure (recipient, purpose, validity window) to the reference, so that a reference is re-resolved when its terms expire [56]. *Attribution* binds a reference to an accountable author or, through anonymous credentials and the pseudonyms the European Digital Identity Wallet must support, to an eligible but unidentified one [57, 58, 44]. *Verifiable credentials* let a reference carry an issuer's attested claim about its source, e.g., that a filing was signed by a member of the bar or that an emissions figure was checked by an accredited verifier [16, 59]. *Capability discovery* shows agents, after each interaction, only the actions currently valid, in the hypermedia style of REST [60] as extended to multi-agent systems [61, 62, 63]; current LLM agents instead receive statically linked tool catalogues [49, 64]. The unifying scientific contribution is a *theory of chosen guarantees*: a formal

account of which NFP bundles an island can provide, at which coordination cost, under which threat model, and for which stakeholders. Neither hypertext research nor Web engineering offers this today (b.2).

Figure 1 shows five example islands within the Web. Each island is a part of the Web's network whose clients and servers raise the contract that the Web offers with the NFPs that their community requires. As an example, *Island L* lets a court or author cite a passage that resolves to exactly the version relied on, and it makes a fabricated citation fail visibly. It leaves a trail of reuse that a copyright dispute, or any scheme that remunerates reuse, can follow, and it makes terms such as a rightholder's reservation against text and data mining [51, 46] verifiable in the reference. It does not, however, establish that a cited source is correct, and it records no reuse that happens outside the island. *Island S* lets suppliers report confidential emission figures that can be checked without decryption; *Island M* lets readers see who stands behind an account and how a post was edited or deleted, and lets researchers obtain exposure statistics; *Island N* keeps signed, content-hashed research claims, as published in the deployed nanopublication network [65, 66, 67], citable and attributed. It ties each claim through span addressing to the specific passage of the source on which it rests. *Island F*, finally, lets fixed installations interact over binary interfaces at low latency while giving up evolvability.

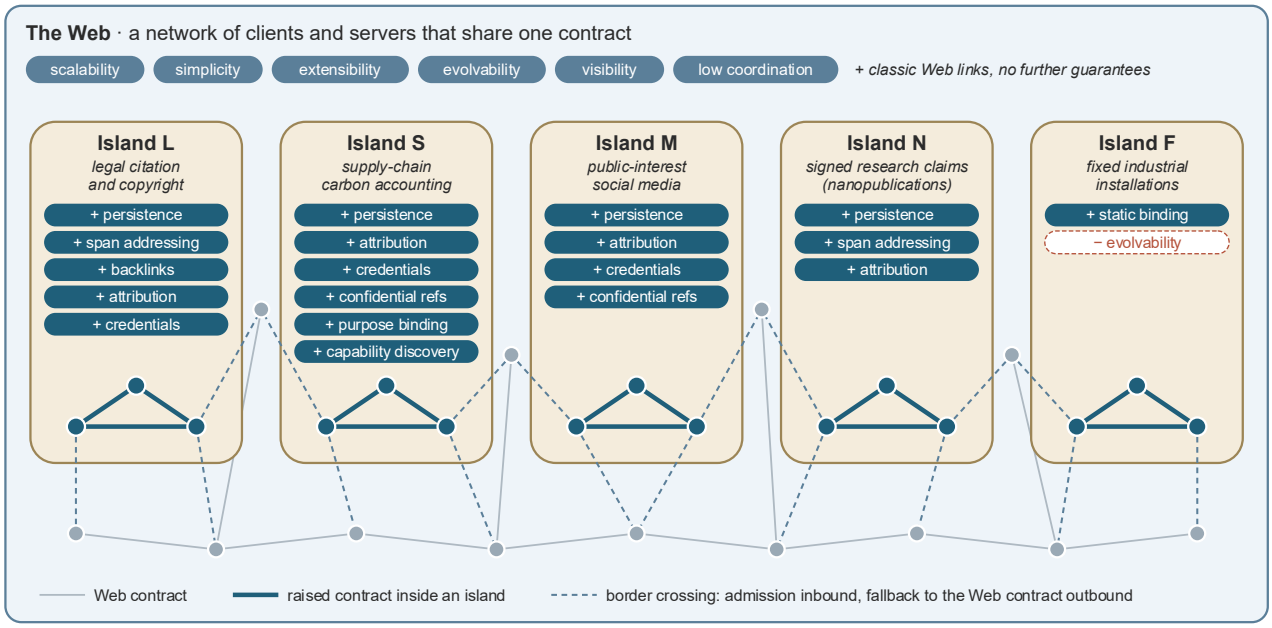


Figure 1. Five example islands within the Web. The Web is a network of clients and servers that share one contract with its own NFPs (top); outside any island, references are classic Web links without further guarantees (thin links). Each island is a part of this network whose members raise the contract by adding NFPs (filled), so that links inside the island carry stronger guarantees (thick links). Island F, the downgrade island, trades evolvability for static binding. Links that cross an island's border pass admission inbound and fall back to the Web contract outbound (dashed). These are the island drafts that the project solidifies and uses as requirements documents for its architecture.

2. State of knowledge and research gap

Hypertext research traditionally assumed rich, span-anchored links (e.g., [52, 68, 69]), and its agenda included versioning and computed structure (e.g., [1]). A link would persist, and it would point at a specific range of bytes rather than a full page. Modifying a link would create a new version instead of replacing the link. Open-hypermedia systems in addition implemented *bidirectional* links and *referential integrity* at workgroup scale [70, 71]; link rot had thus been addressed even before the Web made it endemic. The Web dropped these guarantees to scale [1, 48], and this was a sensible and pragmatic choice at the time.

Since the mid-1990s, when the Internet Archive began to archive the Web [72], remedial infrastructure has kept this architecture and compensated for its drawbacks, such as link rot and misattribution. This infrastructure includes persistent identifier overlays, Web archives, Memento [73], Perma.cc [34], and robust hyperlinks [74], with permanent links such as those of the *Internet Archive's Wayback Machine* or, in the research space, *Zenodo* [75], since link rot undermines reproducibility [29]. Such compensation did not touch the architectural principles of the Web, since changing them might fragment what the W3C's Ethical Web Principles call "one web" [17]. W3C specifications address passages and backlinks on the open Web: *Web Annotation* selectors [76] and *text fragments* [77] identify a passage of a page, and *Webmention* [78] and *Linked Data Notifications* [79], which the W3C *Social Web Working Group* maintains together with *ActivityPub* [80], deliver opt-in backlinks. None of them, however, pins a version or verifies what it points to, since this would induce

architectural change with undesired side effects (e.g., it would undermine scalability of the Web). Dedicated services assemble specific NFP profiles as well, e.g., persistence via *Perma.cc* or *Zenodo*, provenance via *C2PA* [37], and identity and notarization via *DOIs* or *blockchains*. *Solid* [81] brings *data sovereignty*, since users keep their data in personal stores of their choice; *privacy and access control*, since each application reads only what the user grants it; and *portability*, since the data is decoupled from individual applications. Remuneration for reuse followed a similar path. Xanadu tied *transclusion* to payment, so that each delivered fragment would earn its owner a royalty, which Nelson later recast as *transcopyright* [82]. Micropayment schemes for the Web [83] did not take hold, in part because small payments impose mental transaction costs on readers [84]. They now return for machine readers. The status code 402 "Payment Required", reserved in HTTP for decades [85], is used to charge AI crawlers per fetch [86] and agents per request [50]. Cloudflare's Pay Per Use pays publishers for each use of their content that AI companies report themselves [87]. None of these schemes rests on a record of use that both sides can verify. These developments are evidence of the demand for chosen guarantees that extend beyond those of the basic Web architecture. However, each is an individual remedy that provides one fixed profile, and they do not touch the architectural foundation of the Web.

Today's re-decentralization efforts relocate *control* and leave *addressability* as it was. Solid's pods keep location-addressed, mutable resources [81], and data spaces relocate contracts and governance [88]. Federated social media behaves similarly: *ActivityPub* distributes servers but keeps posts location-addressed and mutable [80], and Bluesky's *AT Protocol* signs and content-addresses each user's repository [89] but leaves references between posts unverified and offers one fixed set of guarantees to all communities. In the regulatory space, EU data law follows the same line: the *Data Act* and the portability rights of the *GDPR* and the *DMA* move *copies of data* between holders [45]. The Solid deployments that implement them (cf. our own [90, 91]) leave references location-addressed. However, raw-data transfer cannot sustain a market, hence we have recently seen proposals of trading *purpose-bound data derivations* whose terms limit their shelf life and use [56]. There is today no reference layer that could name such a derivation, pin it to a version, or carry its terms. *Usage control* formalized such terms [92, 93], and *sticky policies* bind them to the data [94]. Enforcement after release, however, requires a trusted monitor at the recipient or falls back on detection after the fact [93]. RFC 9518 explains why open specifications alone do not prevent centralization, and we adopt the design-level mitigations it lists [11]. It is relevant to note that network architecture faced the same question one layer below. Clark et al. argued that an Internet embedded in society must leave room for the *tussle* between stakeholders with conflicting interests, and proposed design for change and controlled transparency as its tenets [95, 96]. Plutarch argued for heterogeneous network contexts joined by translating functions at their borders [97]. These positions state principles on the network layer that mirror the issues we highlight on the application layer of the World Wide Web, but they concern packets and network addresses rather than references and information resources.

As a community, we have thus built imperfect remedies instead of an architected approach. The reason is that an architected approach is hard: it requires consolidating fields that have so far advanced separately, namely hypertext, distributed systems, cryptography, multi-agent systems, and law.

The methods that make such a shift possible today (Section 1) were not available in the early days of the Web: *Content-addressed* and *versioned storage* (Git; IPFS [42]; Software Heritage identifiers [98]) and *provenance attestation* [37, 99] now supply, at Web scale, the substrate that Xanadu [52, 53] lacked. ISLANDS takes from Xanadu its traceability of reuse, not control over reading: a quoted passage stays linked to its source, but nothing restricts copying or enforces terms on readers' devices, which separates *islands* from digital rights management. They are, however, storage and signing primitives without span semantics, a link layer, or a governance model. *Nanopublications* [65, 66] form a deployed, federated network of immutable, signed semantic statements with self-verifying identifiers. They do not, however, offer span addressing over arbitrary content, *transclusion*, or confidentiality-preserving resolution. From the perspective of this proposal, nanopublications may become an *island* at statement granularity (Island N in Figure 1). With respect to who may see, trust, and act on the Web, too, the building blocks exist but are not part of the Web's architecture. With respect to confidentiality, communities need results computed over data that no single member may disclose. Secure multi-party computation (SMPC) has run in production since 2008 [54] and general-purpose protocols and implementations such as SPDZ [100] and MP-SPDZ [43] make computation practical. Our PEP-LCA protocol computes joint supply-chain assessments across up to 50,000 participants without a trusted third party [101], and a follow-on system anchors share disclosure in Solid's access-control layer [102]. Again, each of these deployments is, however, an individual application, and none exposes its result as a reference that others can cite and re-resolve. Confidentiality keeps data from the parties that use it; trust needs the converse, a verifiable statement about who stands behind the data. Verifiable credentials address this part of the problem. The W3C data model [16], selective-disclosure formats such as SD-JWT [103], and the European Digital

Identity Wallet [44] let a holder prove an issuer's claim (e.g., a professional license) without revealing more than needed [59, 104]. They are, however, presented to one verifier within one session, and they are not bound to the references that the claim concerns. Credentials tell a client whom it deals with; autonomous agents also need to know what they may do. Hypermedia multi-agent systems research argues for run-time *affordance* discovery [61, 105]. Current LLM-agent practice, in contrast, uses statically linked tool lists [49, 64], which indirect prompt injection turns into an attack surface [106], and no existing design connects *capability discovery* to a verifiable reference layer.

What we do not know. Today, we do not know how different NFP selections work together. Each hypertext system and each remedy above fixes one bundle of guarantees by design. None of them tells us which bundles are technically feasible in the first place, at what coordination cost, and against which threat model. It is open, for instance, whether *confidentiality-preserving references* can be combined with bidirectional backlinks without the backlinks revealing what the references hide, or whether attribution and anonymity can coexist in one island. This is a major gap. O1 and WP1 determine which bundles society requires, and O2 and WP2 determine which of them are attainable, at what cost, and under which threat model.

The gap. We are not aware of any system that composes immutability, span-level verifiable addressing, transclusion, confidentiality-preserving resolution, and agent-facing affordance discovery *at the architectural level*. Closing this gap requires an understanding of each of these domains, spanning distributed computing, decentralization, and the societal requirements behind them. It must also be done in a backwards-compatible way that reuses the Web's architecture as deployed today and accepts its NFPs in bounded subsystems. Finally, for practical adoption, any such architecture needs a theory that predicts which NFP bundles fit a given deployment context and community.

3. Objectives

ISLANDS asks one research question: *which guarantees can a Web architecture let communities choose, and at what cost?* Its goal is to answer this question for the communities that need such guarantees and to demonstrate the answer in practice. Answering it closes the gap stated above, so that society no longer pays piecemeal for guarantees that the architecture lacks. This involves:

- **O1 — Societally required islands.** We establish which *islands*, i.e., which bundles of NFPs, society requires today and in the foreseeable future. We start from the five drafts of Figure 1: Islands L, S, M, N, and F. We will consider, at the time of the project, further possible islands, including one that supports the requirements of collaborating heterogeneous artificial agents [63]. An *architectural decision record* states which islands the project does not pursue.
- **O2 — Theory of chosen guarantees.** Next, we isolate the *link classes* that the required islands need and develop a formal model that relates them to coordination and economic costs, island size, threat models, and stakeholder incentives. We foresee this taxonomy to include a persistence and integrity ladder (*Classic Jumplinks*, *Archived*, *Trusted*, *Verified*, and *Cryptographic Links*, in increasing strength) and at least five cross-cutting axes (attribution, trust, usage, modality, *link time*). For each class, we prove what it minimally requires, and we derive decision procedures that an island designer can apply.
- **O3 — Border translation.** We determine how NFPs, and the concepts that support them (versions, identities, credentials, usage terms, encodings), are translated when information enters or leaves an island: which references are upgraded at admission, which are downgraded on exit, and how both happen without the silent loss of a guarantee.
- **O4 — Island technologies.** We build the technologies that the required islands need. This will, minimally, include a reference and *transclusion* layer over content-addressed storage; confidential references that turn SMPC-based resolution into an addressing primitive; and an agent layer with state-dependent *capability discovery* and design-time *Static Links*.
- **O5 — Validation and incentives.** We instantiate and evaluate Islands L, S, and M with their communities and Island F in industrial installations, and we measure whether the islands change each actor's incentive to centralize [11]. We assess the technical artifacts of each island from business, legal, and economic perspectives: which business models can sustain the island, whether its guarantees meet the law of its domain, and how it shifts costs and incentives.

4. Research strategy and methods

The proposed project's work packages follow its objectives: WP1 establishes the *islands* that society requires. WP2 and WP3 then derive, in parallel, what these islands need inside and at their borders. WP4 builds the technologies, and WP5, which starts in parallel with WP1, validates them with the respective communities.

The overall approach is design-science research in annual cycles, and three choices make an ambitious goal realistic. First, we do not start from an empty design space but from the five island drafts of Figure 1, each backed by a community that WP5 involves from M12 onward. Second, the technical building blocks are proven. Our PEP-LCA protocol computes joint assessments across up to 50,000 participants [101], and the PI's Web of Things descriptions are W3C Recommendations deployed in industrial products [107]. Our *signifier*-based agents reason over run-time action repertoires [105]. These systems de-risk the project; they are not its contribution. What ISLANDS adds is new in kind: guarantees that communities choose as an architectural principle of the Web, and a theory that predicts which guarantees an island can hold. Third, that theory is tested against working systems once a year. Each integration milestone records which design decisions the calculus predicted and which forced a revision, so that a wrong prediction surfaces early as a result and not late as a failure.

WP1 — Societally required islands (M1–M18). Starting from the five drafts in Figure 1, we determine with affected stakeholders which NFP bundles their domain requires. The drafts cover legal citation and copyright (Island L), supply-chain carbon accounting (Island S), public-interest social media (Island M), the nanopublication record (Island N), and fixed industrial installations (Island F). Interviews with each class of actor, the RFC 9518 checklist [11], and the W3C TAG's societal-impact self-review [108] test each draft. Business, legal, and economic analyses ask which business model could sustain each island, which legal obligations its NFPs must meet, and which incentives keep or erode each guarantee. WP1 delivers an account of the required islands and an *architectural decision record* of the islands that ISLANDS chooses not to pursue further. WP1 also searches for islands that the five drafts do not cover, specifically one for collaborating heterogeneous artificial agents [63], whose members differ in architecture and need to discover what the others can do.

WP2 — Link classes and the theory of chosen guarantees (M6–M60). WP2 isolates the *link classes* that these islands need, from *Classic Jumplinks* to *Cryptographic Links*, with cross-cutting axes for attribution, trust, usage, modality, and *link time*. It formalizes them as a *guarantee calculus* in the information-flow control tradition. Each class carries two proof obligations, *realizability* and *necessity*, so that the calculus predicts which classes an island of given size, stake, and threat model can sustain, and at what cost.

WP3 — Border checks (M6–M48, in parallel with WP2). WP3 determines how NFPs, and the concepts that support them, are translated when information enters or leaves an island. It establishes which references are upgraded at admission, which are downgraded on exit, and which cannot cross. Its governing rule is that no reference loses a guarantee silently. WP3 delivers a translation table, an *admission protocol* based on *verifiable credentials* [16], and border gateways for each island.

WP4 — Island technologies (M12–M54). From the islands and their link classes, we build the technologies they require. These are a reference and *transclusion* layer over content-addressed storage (T4.1), confidential references that generalize our SMPC system [101, 102] into an addressing primitive (T4.2), and an agent layer with state-dependent *capability discovery* and design-time *Static Links* (T4.3).

WP5 — Validation islands and incentives (M12–M60). From M12, each island's community joins the project, e.g., legal editors, supply-chain partners, public-interest publishers, research infrastructures, and installation operators. Each community reviews the WP1 account of its island before the technologies are built. We then instantiate Islands L, S, and M with their communities and demonstrate Island F in industrial installations. We measure which guarantees each island holds and at what cost, and we repeat the WP1 interviews to test whether the islands changed each actor's incentive to centralize. Finally, we assess the deployed artifacts from business, legal, and economic perspectives.

Integration and openness. Annual integration milestones bind the prototypes into one architecture and revise the WP1 account and the calculus. All software and publications are open, and all specifications go to the W3C groups that maintain the protocols islands build on. Each work package thus narrows the gap of Section 2: together, they replace piecemeal fixes with guarantees that communities choose.

5. Team

The research question cannot be answered from within one field: it needs a team that covers Web architecture, ubiquitous computing, human-computer interaction, and law, and that runs systems in these fields. The PI has built all four research lines, and his group is such a team. In *Web architecture*, his work on the Web of Things contributed to a W3C standard [107, 109], and his work on hypermedia multi-agent systems introduced *signifiers* through which agents discover what they can do [61, 62, 105]. *Ubiquitous computing* extends *islands* from documents to devices and places [110, 111]. *Human-computer interaction* informs how clients and agents are shown the obligations they accept when they cross an island's border [63]. In *law*, regulation expressed as technical architecture [24, 112, 113] and platform auditing under the DSA [18, 19] ground Islands L and M.

Each line has so far addressed one part of the problem; the *theory of chosen guarantees* needs all of them, because an island's guarantees are at once architectural, experienced by people, and legally consequential. Two members of the PI's group join the project with the systems they built: **Luka Bekavac** with the instruments for platform auditing [18, 19], and **Jan Grau** with the PEP-LCA SMPC system and its Solid integration [101, 102]. The PI stepped down as Dean in 2026 to pursue this programme (≥40% commitment).

6. Ambition, risk, gain

ISLANDS tests whether a coordination cost that the Web's designers judged prohibitive [3, 4] is sustainable at *island* scale with today's technologies and approaches. It also tests whether cryptographic reference resolution can reach interactive latency, and whether actors with divergent incentives will adopt shared guarantees. Finally, it tests whether, for the communities that need them, guarantees built into the architecture cost society less than today's platform-level fixes. These are risky hypotheses, hence O1 asks which guarantees society requires, O2 treats them as *chosen*, and O5 measures incentives alongside latency. If our hypotheses hold, we will deliver a validated architecture for the parts of the information space where the Web's founding trade-off now fails at measurable cost. These are chiefly the parliamentary, legal, and scientific record, public discourse on social media, and regulated industrial data. Autonomous agents could then act on verifiable information in these spaces. If our hypotheses fail, the theory still delimits which guarantees are attainable, for whom, and at what cost.

The alternative is to continue as today, with piecemeal fixes whose costs society absorbs one incident at a time. Where this leads is sketched, in exaggerated form, by the *Dead Internet Theory*, which holds that most online content and activity is already produced by bots [114]. We do not endorse the theory, but its premise is moving closer: automated traffic surpassed human traffic on the Web in 2024 [48], and language models degrade when trained on content that models generated [115]. A Web that cannot state who stands behind a document, which version a claim rests on, or whether a text was generated, becomes a space in which neither people nor machines can rely on what they read. Courts, parliaments, and markets would then have to rebuild trust outside the Web, at far greater cost than the coordination the Web avoided. ISLANDS offers the architectural alternative: guarantees that communities choose and that hold where they are needed.

References (not counted towards the page limit)

[1] F. G. Halasz. Reflections on NoteCards: Seven issues for the next generation of hypermedia systems. *Communications of the ACM* 31(7):836–852, 1988. doi:10.1145/48511.48514

[2] F. Halasz, M. Schwartz. The Dexter hypertext reference model. *Communications of the ACM* 37(2):30–39, 1994. doi:10.1145/175235.175237

[3] T. Berners-Lee. *Information Management: A Proposal.* CERN, March 1989.

[4] R. T. Fielding, R. N. Taylor. Principled design of the modern Web architecture. *ACM Transactions on Internet Technology* 2(2):115–150, 2002. doi:10.1145/514183.514185

[5] T. Gillespie. The politics of 'platforms'. *New Media & Society* 12(3):347–364, 2010. doi:10.1177/1461444809342738

[6] J.-C. Plantin, C. Lagoze, P. N. Edwards, C. Sandvig. Infrastructure studies meet platform studies in the age of Google and Facebook. *New Media & Society* 20(1):293–310, 2018. doi:10.1177/1461444816661553

[7] J. van Dijck, T. Poell, M. de Waal. *The Platform Society: Public Values in a Connective World.* Oxford University Press, 2018. doi:10.1093/oso/9780190889760.001.0001

[8] A. Helmond. The platformization of the Web: Making Web data platform ready. *Social Media + Society* 1(2), 2015. doi:10.1177/2056305115603080

[9] L. Bekavac, S. Mayer. Broken links and fading stories: How dominant online platforms subvert the hypermedia foundation of the Web. *ACM Web Science Conference (WebSci '26)*, 221–226, 2026. doi:10.1145/3795766.3799767

[10] T. Berners-Lee. Long live the Web. *Scientific American* 303(6):80–85, 2010. doi:10.1038/scientificamerican1210-80

[11] M. Nottingham. *Centralization, Decentralization, and Internet Standards.* RFC 9518, IETF, 2023.

[12] F. Huszár, S. I. Ktena, C. O'Brien, L. Belli, A. Schlaikjer, M. Hardt. Algorithmic amplification of politics on Twitter. *PNAS* 119(1):e2025334119, 2022. doi:10.1073/pnas.2025334119

[13] Regulation (EU) 2022/2065 (Digital Services Act), Arts. 34, 35(1)(k), and 40; Commission Delegated Regulation (EU) 2025/2050 on data access for vetted researchers.

[14] C. Goanta, S. Zannettou, R. Kaushal, J. van de Kerkhof, T. Bertaglia, T. Annabell, H. Gui, G. Spanakis, A. Iamnitchi. The great data standoff: Researchers vs. platforms under the Digital Services Act. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)* 20(1):874–888, 2026. doi:10.1609/icwsm.v20i1.42671

[15] European Commission. Commission fines X €120 million under the Digital Services Act. Press release, 5 December 2025.

[16] M. Sporny, T. Thibodeau Jr, I. Herman, G. Cohen, M. B. Jones (eds.). *Verifiable Credentials Data Model v2.0.* W3C Recommendation, 15 May 2025.

[17] D. Appelquist, H. Beeman, A. Guy (eds.). *Ethical Web Principles.* W3C Statement, 12 December 2024.

[18] L. Bekavac, J. Strecker-Bischoff, K. Garcia, S. Mayer, A. Tamò-Larrieux. Scrutinizing systemic risks in personalized recommender systems through sock-puppet auditing of VLOPs. *ACM Transactions on Recommender Systems* 5(1):1–47, 2026. doi:10.1145/3795516

[19] L. Bekavac, S. Mayer. Platforms' research API data access: What users see vs. what researchers can retrieve. *ACM FAccT* 2026. doi:10.1145/3805689.3812237

[20] M. Schneider. Updating DSA transparency reports: How TikTok's silent correction tests Implementing Regulation. Open Terms Archive, 28 August 2026. opentermsarchive.org

[21] Cour d'appel de Paris, rulings of 29 October 2026 upholding the July 2026 order of the Autorité de la concurrence against Meta (neighbouring rights of press publishers); reported by AFP, "French court rules Meta must negotiate over rights payments", France 24, 29 October 2026. **[replace with the decision reference once published]**

[22] L. Lessig. *Code and Other Laws of Cyberspace.* Basic Books, 1999.

[23] M. Hildebrandt. *Smart Technologies and the End(s) of Law.* Edward Elgar, 2015. doi:10.4337/9781849808774

[24] A. Tamò-Larrieux, S. Mayer, Z. Zihlmann. Not hardcoding but softcoding data protection. *Technology and Regulation* 2021:17–34, 2021. doi:10.71265/a1wsy445

[25] L. Zard. Internet ozone layer: A vision for reclaiming human freedom in information civilization. Working paper, SSRN 5735624, 2025.

[26] J. Yasskin, S. Capadisli (eds.). *User Agents.* W3C Technical Architecture Group, Group Note, 22 October 2026.

[27] Reddit, Inc. Q3 2026 earnings call, 29 October 2026 (transcript: The Motley Fool, 30 October 2026); reported as "As search traffic dries up, Reddit announces it's shifting to a logged-in experience", TechCrunch, 29 October 2026.

[28] A. Spencer, J. Sheridan, D. Thomas, D. Pullinger. UK government Web continuity: Persisting access through aligning infrastructures. *International Journal of Digital Curation* 4(1):107–124, 2009. doi:10.2218/ijdc.v4i1.82

[29] M. Klein, H. Van de Sompel, R. Sanderson, H. Shankar, L. Balakireva, K. Zhou, R. Tobin. Scholarly context not found: One in five articles suffers from reference rot. *PLoS ONE* 9(12):e115253, 2014. doi:10.1371/journal.pone.0115253

[30] L. de Nadal, P. Jančárik. Beyond the deepfake hype: AI, democracy, and "the Slovak case". *Harvard Kennedy School Misinformation Review* 5(4), 2024. doi:10.37016/mr-2020-153

[31] Regulation (EU) 2023/1542 (Batteries Regulation), Art. 77 and Annex XIII (battery passport, applicable from 18 February 2027).

[32] Council conclusions inviting the introduction of the European Case Law Identifier (ECLI) and a minimum set of uniform metadata for case law. OJ C 127, 29 April 2011.

[33] Council conclusions inviting the introduction of the European Legislation Identifier (ELI). OJ C 325, 26 October 2012.

[34] J. Zittrain, K. Albert, L. Lessig. Perma: Scoping and addressing the problem of link and reference rot in legal citations. *Legal Information Management* 14(2):88–99, 2014. doi:10.1017/S1472669614000255

[35] D. Charlotin. *AI Hallucination Cases* database, damiencharlotin.com/hallucinations, accessed May 2026. **[re-check counts at submission]**

[36] *R (Ayinde) v. London Borough of Haringey; Al-Haroun v. Qatar National Bank*
[2025] EWHC 1383 (Admin), Divisional Court, 6 June 2025.

[37] Coalition for Content Provenance and Authenticity. *C2PA Technical Specification*, v2, 2024–2025.

[38] Regulation (EU) 2024/1689 (Artificial Intelligence Act), Art. 50.

[39] Landgericht München I, judgment of 11 November 2025, 42 O 14139/24 (*GEMA v. OpenAI*), not final. **[re-check appeal status at submission]**

[40] *The New York Times Co. v. Microsoft Corp. and OpenAI*, No. 1:23-cv-11195 (S.D.N.Y., filed 27 December 2023).

[41] George Washington University Law School. *Database of AI Litigation (DAIL)*, accessed 2026. **[re-check at submission]**

[42] J. Benet. IPFS: Content addressed, versioned, P2P file system. arXiv:1407.3561, 2014.

[43] M. Keller. MP-SPDZ: A versatile framework for multi-party computation. *ACM CCS 2020*, 1575–1590. doi:10.1145/3372297.3417872

[44] Regulation (EU) 2024/1183 amending Regulation (EU) No 910/2014 as regards establishing the European Digital Identity Framework.

[45] Regulation (EU) 2023/2854 (Data Act); Regulation (EU) 2016/679 (GDPR), Art. 20; Regulation (EU) 2022/1925 (Digital Markets Act), Art. 6(9).

[46] Regulation (EU) 2024/1689 (AI Act), Art. 53(1)(c).

[47] Regulation (EU) 2023/956 (Carbon Border Adjustment Mechanism), Arts. 8, 18, and 35.

[48] Imperva (Thales). *2025 Bad Bot Report.* April 2025.

[49] *Model Context Protocol Specification*, version 2025-06-18. modelcontextprotocol.io.

[50] Coinbase Developer Platform. Introducing x402: A new standard for internet-native payments. 6 May 2025.

[51] Directive (EU) 2019/790 (Copyright in the Digital Single Market), Art. 4(3).

[52] T. H. Nelson. *Literary Machines.* Mindful Press, 1981.

[53] T. H. Nelson. Xanalogical structure, needed now more than ever. *ACM Computing Surveys* 31(4es), 1999.

[54] P. Bogetoft, D. L. Christensen, I. Damgård, M. Geisler, T. Jakobsen, M. Krøigaard, et al. Secure multiparty computation goes live. *Financial Cryptography and Data Security 2009*, LNCS 5628:325–343. doi:10.1007/978-3-642-03549-4_20

[55] D. Bogdanov, L. Kamm, B. Kubo, R. Rebane, V. Sokk, R. Talviste. Students and taxes: A privacy-preserving study using secure computation. *Proceedings on Privacy Enhancing Technologies* 2016(3):117–135. doi:10.1515/popets-2016-0019

[56] R. Verborgh. Inside the insight economy. ruben.verborgh.org, 12 August 2025.

[57] D. Chaum. Security without identification: Transaction systems to make big brother obsolete. *Communications of the ACM* 28(10):1030–1044, 1985. doi:10.1145/4372.4373

[58] J. Camenisch, A. Lysyanskaya. An efficient system for non-transferable anonymous credentials with optional anonymity revocation. *EUROCRYPT 2001*, LNCS 2045:93–118. doi:10.1007/3-540-44987-6_7

[59] J. Sedlmeir, R. Smethurst, A. Rieger, G. Fridgen. Digital identities and verifiable credentials. *Business & Information Systems Engineering* 63(5):603–613, 2021. doi:10.1007/s12599-021-00722-y

[60] R. T. Fielding. *Architectural Styles and the Design of Network-based Software Architectures.* PhD thesis, University of California, Irvine, 2000.

[61] A. Ciortea, S. Mayer, F. Gandon, O. Boissier, A. Ricci, A. Zimmermann. A decade in hindsight: The missing bridge between multi-agent systems and the World Wide Web. *AAMAS 2019*, 1659–1663.

[62] D. Vachtsevanou, A. Ciortea, S. Mayer, J. Lemée. Signifiers as a first-class abstraction in hypermedia multi-agent systems. *AAMAS 2023*, 1200–1208.

[63] J. Lemée, D. Vachtsevanou, S. Mayer, A. Ciortea. Signifiers for conveying and exploiting affordances: From human-computer interaction to multi-agent systems. *Annals of Mathematics and Artificial Intelligence* 92(4):815–835, 2024. doi:10.1007/s10472-024-09938-6

[64] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, E. Hambro, L. Zettlemoyer, N. Cancedda, T. Scialom. Toolformer: Language models can teach themselves to use tools. *NeurIPS 2023*, 68539–68551.

[65] T. Kuhn, M. Dumontier. Trusty URIs: Verifiable, immutable, and permanent digital artifacts for linked data. *ESWC 2014*, LNCS 8465:395–410. doi:10.1007/978-3-319-07443-6_27

[66] T. Kuhn, C. Chichester, M. Krauthammer, N. Queralt-Rosinach, R. Verborgh, et al. Decentralized provenance-aware publishing with nanopublications. *PeerJ Computer Science* 2:e78, 2016.

[67] Nanopublication Association. *Nanopublications.* nanopub.net, accessed 5 November 2026.

[68] V. Bush. As we may think. *The Atlantic Monthly* 176(1):101–108, 1945.

[69] D. C. Engelbart, W. K. English. A research center for augmenting human intellect. *AFIPS Fall Joint Computer Conference*, 395–410, 1968. doi:10.1145/1476589.1476645

[70] A. M. Fountain, W. Hall, I. Heath, H. C. Davis. MICROCOSM: An open model for hypermedia with dynamic linking. *ECHT '90*, Cambridge University Press, 298–311, 1990.

[71] H. C. Davis. Referential integrity of links in open hypermedia systems. *ACM Hypertext '98*, 207–216, 1998. doi:10.1145/276627.276650

[72] B. Kahle. Preserving the Internet. *Scientific American* 276(3):82–83, 1997. doi:10.1038/scientificamerican0397-82

[73] H. Van de Sompel, M. Nelson, R. Sanderson. *HTTP Framework for Time-Based Access to Resource States: Memento.* RFC 7089, IETF, 2013.

[74] T. A. Phelps, R. Wilensky. Robust intra-document locations. *Computer Networks* 33(1–6):105–118, 2000. doi:10.1016/S1389-1286(00)00043-8

[75] European Organization for Nuclear Research, OpenAIRE. *Zenodo.* CERN, 2013. doi:10.25495/7GXK-RD71

[76] R. Sanderson, P. Ciccarese, B. Young (eds.). *Web Annotation Data Model.* W3C Recommendation, 23 February 2017.

[77] N. Burris, D. Bokan (eds.). *URL Fragment Text Directives.* W3C WICG Draft Community Group Report, 2023.

[78] A. Parecki (ed.). *Webmention.* W3C Recommendation, 12 January 2017.

[79] S. Capadisli, A. Guy (eds.). *Linked Data Notifications.* W3C Recommendation, 2 May 2017.

[80] C. Lemmer-Webber, J. Tallon, E. Shepherd, A. Guy, E. Prodromou. *ActivityPub.* W3C Recommendation, 23 January 2018.

[81] A. V. Sambra, E. Mansour, S. Hawke, M. Zereba, N. Greco, A. Ghanem, D. Zagidulin, A. Aboulnaga, T. Berners-Lee. Solid: A platform for decentralized social applications based on linked data. Technical report, MIT CSAIL and QCRI, 2016.

[82] T. H. Nelson. Transcopyright: A simple legal arrangement for sharing, re-use, and republication of copyrighted material on the net. *Worldwide Computing and Its Applications (WWCA '97)*, LNCS 1274, 7–14, 1997. doi:10.1007/3-540-63343-X_34

[83] R. L. Rivest, A. Shamir. PayWord and MicroMint: Two simple micropayment schemes. *Security Protocols*, LNCS 1189, 69–87, 1997. doi:10.1007/3-540-62494-5_6

[84] N. Szabo. Micropayments and mental transaction costs. *2nd Berlin Internet Economics Workshop*, May 1999.

[85] R. Fielding, M. Nottingham, J. Reschke (eds.). *HTTP Semantics.* RFC 9110, IETF, 2022, §15.5.3.

[86] Cloudflare. Introducing pay per crawl: Enabling content owners to charge AI crawlers for access. Cloudflare blog, 1 July 2025.

[87] R. Teixeira, J. Galilee, K. Johnston, S. Else. Pay Per Use: When AI uses your work, you should get paid. Cloudflare blog, 30 September 2026.

[88] B. Otto, M. ten Hompel, S. Wrobel (eds.). *Designing Data Spaces: The Ecosystem Approach to Competitive Advantage.* Springer, 2022. doi:10.1007/978-3-030-93975-5

[89] M. Kleppmann, P. Frazee, J. Gold, J. Graber, D. Holmgren, D. Ivy, J. Johnson, B. Newbold, J. Volpert. Bluesky and the AT Protocol: Usable decentralized social media. *ACM CoNEXT Workshop on the Decentralization of the Internet (DIN '24)*, 1–7, 2024. doi:10.1145/3694809.3700740

[90] L. Bekavac. From data portability to data control: Making regulation technically real. *4th Solid Symposium*, London, 2026.

[91] L. K. Seiling, J. Stein, S. Mayer, L. Bekavac. A Solid research infrastructure leveraging DMA data portability. 2026. **[TODO: venue]**

[92] J. Park, R. Sandhu. The UCON_ABC usage control model. *ACM Transactions on Information and System Security* 7(1):128–174, 2004. doi:10.1145/984334.984339

[93] A. Pretschner, M. Hilty, D. Basin. Distributed usage control. *Communications of the ACM* 49(9):39–44, 2006. doi:10.1145/1151030.1151053

[94] S. Pearson, M. Casassa-Mont. Sticky policies: An approach for managing privacy across multiple parties. *IEEE Computer* 44(9):60–68, 2011. doi:10.1109/MC.2011.225

[95] D. D. Clark, J. Wroclawski, K. R. Sollins, R. Braden. Tussle in cyberspace: Defining tomorrow's Internet. *IEEE/ACM Transactions on Networking* 13(3):462–475, 2005. doi:10.1109/TNET.2005.850224

[96] D. D. Clark, K. Sollins, J. Wroclawski, T. Faber. Addressing reality: An architectural response to real-world demands on the evolving Internet. *ACM SIGCOMM FDNA '03*, 247–257, 2003. doi:10.1145/944759.944761

[97] J. Crowcroft, S. Hand, R. Mortier, T. Roscoe, A. Warfield. Plutarch: An argument for network pluralism. *ACM SIGCOMM FDNA '03*, 258–266, 2003. doi:10.1145/944759.944763

[98] ISO/IEC 18670:2025. *Information technology: SoftWare Hash IDentifier (SWHID) specification.*

[99] L. Moreau, P. Missier (eds.). *PROV-DM: The PROV Data Model.* W3C Recommendation, 2013.

[100] I. Damgård, V. Pastro, N. Smart, S. Zakarias. Multiparty computation from somewhat homomorphic encryption. *CRYPTO 2012*, LNCS 7417:643–662. doi:10.1007/978-3-642-32009-5_38

[101] J. Grau. Towards decentralized privacy-enhancing life cycle assessment. MSc thesis, University of St. Gallen, 2025.

[102] J. Grau, M. Maftun, K. Garcia, S. Mayer. Solid and secure multi-party computation for a circular economy. *Solid Symposium* 2026.

[103] D. Fett, K. Yasuda, B. Campbell. *Selective Disclosure for JSON Web Tokens.* RFC 9901, IETF, 2025.

[104] A. Mühle, A. Grüner, T. Gayvoronskaya, C. Meinel. A survey on essential components of a self-sovereign identity. *Computer Science Review* 30:80–86, 2018. doi:10.1016/j.cosrev.2018.10.002

[105] D. Vachtsevanou, B. de Lima, A. Ciortea, J. F. Hübner, S. Mayer, J. Lemée. Enabling BDI agents to reason on a dynamic action repertoire in hypermedia environments. *AAMAS 2024*, 1856–1864.

[106] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, M. Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. *ACM AISec '23*, 79–90, 2023. doi:10.1145/3605764.3623985

[107] S. Kaebisch, M. McCool, E. Korkan (eds.). *Web of Things (WoT) Thing Description 1.1.* W3C Recommendation, 2023.

[108] D. A. Norman. *The Design of Everyday Things.* Revised and expanded edition, Basic Books, 2013.

[109] S. Mayer. *Interacting with the Web of Things.* Doctoral thesis, ETH Zurich, 2014.

[110] L. Bekavac, S. Mayer, J. Strecker. QR-code integrity by design. *Extended Abstracts of CHI 2024*, 1–9. doi:10.1145/3613905.3651006

[111] J. Strecker, S. Mayer, K. Bektaş. Towards societally beneficial personalized realities: A conceptual foundation for responsible ubiquitous personalization systems. *ACM DIS 2025*, 1792–1814. doi:10.1145/3715336.3735709

[112] A. Tamò-Larrieux, C. Guitton, S. Mayer. A computational science perspective on the legal system. *Nature Computational Science* 5(11):990–991, 2025. doi:10.1038/s43588-025-00827-4

[113] C. Guitton, A. Tamò-Larrieux, S. Mayer. *AI and Law: How Automation is Changing the Law.* Routledge, 2025.

[114] Y. Walter. Artificial influencers and the dead internet theory. *AI & Society* 40:239–240, 2025 (online 2024). doi:10.1007/s00146-023-01857-0

[115] I. Shumailov, Z. Shumaylov, Y. Zhao, N. Papernot, R. Anderson, Y. Gal. AI models collapse when trained on recursively generated data. *Nature* 631:755–759, 2024. doi:10.1038/s41586-024-07566-y

Section b: Curriculum Vitae and Track Record

1. Personal details

Prof. Dr. Simon Mayer; born 16 July 1987; Austrian. ORCID 0000-0001-6367-3454.

Education. Dr.sc. ETH in Computer Science, ETH Zurich (defense 12 September 2014; 12.3 years before 1 January 2027, within the CoG window of 5–15 years). M.Sc. ETH in Computer Science (2010); B.A. in Economics and Business Administration, University of Zurich (2014); MAS in Secondary and Higher Education, ETH Zurich (2014).

Current position. Since 2018: Full Professor of Computer Science (Interaction- and Communication-based Systems), University of St. Gallen (USG), one of the four founding professors of its School of Computer Science; Director, Institute of Computer Science. Since 2020: Board Member, Research Center for Information Law, USG.

Previous positions. 2024–02/2026: Dean, School of Computer Science, USG (2022–2024 Academic Director; 2018–2024 Managing Director, Institute of Computer Science). 2023: Visiting Professor, University of Bologna. 2018–2020: Senior Scientist, Distributed Systems Group, ETH Zurich. 2017–2018: Area Manager "Cognitive Products", Pro2Future GmbH, and Project Scientist, Graz University of Technology. 2014–2017: Research Scientist, Senior Researcher, and Senior Key Expert, Siemens Corporate Technology, Berkeley, USA. 2015–2017: Visiting Lecturer, UC Berkeley and ETH Zurich. 2013: Visiting Researcher, Ghent University. 2012: Visiting Researcher, MIT.

2. Research achievements and peer recognition

Research achievements (most recent first within each line).

1. Bekavac, **Mayer**. *Broken Links and Fading Stories: How Dominant Online Platforms Subvert the Hypermedia Foundation of the Web.* ACM WebSci 2026.
2. Bekavac, **Mayer**. *Platforms' Research API Data Access: What Users See vs. What Researchers Can Retrieve.* ACM FAccT 2026; and Bekavac, Strecker-Bischoff, Garcia, **Mayer**, Tamò-Larrieux. *Scrutinizing Systemic Risks in Personalized Recommender Systems Through Sock-Puppet Auditing of VLOPs.* ACM Transactions on Recommender Systems 5(1), 2026.
3. Grau, Maftun, Garcia, **Mayer**. *Solid and Secure Multi-Party Computation for a Circular Economy.* Solid Symposium 2026, building on Grau's MSc thesis (2025, supervised by the PI with Garcia, Mitrokotsa, and Cheng).
4. Guitton, Tamò-Larrieux, **Mayer**. *AI and Law: How Automation Is Changing the Law.* Routledge, 2025.
5. Tamò-Larrieux, Guitton, **Mayer**. *A Computational Science Perspective on the Legal System.* Nature Computational Science, 2025.
6. Vachtsevanou, Ciortea, **Mayer**, Lemée. *Signifiers as a First-class Abstraction in Hypermedia Multi-Agent Systems.* AAMAS 2023; and Vachtsevanou et al. *Enabling BDI Agents to Reason on a Dynamic Action Repertoire in Hypermedia Environments.* AAMAS 2024.
7. **Mayer**, Fuchs, Brügger, Lian, Ciortea. *Improving Customer Decisions in Web-based E-commerce through Guerrilla Modding.* Nature Machine Intelligence 3(12), 2021.
8. Tamò-Larrieux, **Mayer**, Zihlmann. *Not Hardcoding but Softcoding Data Protection.* Technology and Regulation, 2021.
9. Ciortea, **Mayer**, Gandon, Boissier, Ricci, Zimmermann. *A Decade in Hindsight: The Missing Bridge Between Multi-Agent Systems and the World Wide Web.* AAMAS 2019.
10. **W3C Web of Things Thing Description.** The PI's doctoral and Siemens research contributed to the basis of the W3C Thing Description, which is deployed in industrial products (e.g., Siemens Desigo building management).

Peer recognition. General Chair, Solid Symposium 2027 and ACM Symposium on Computer Science and Law (CS&Law) 2028. Member of the European oversight board on Very Large Online Platform auditing. Associate Editor, PACM IMWUT (since 2021); Editorial Board, IEEE Pervasive Computing (since 2019); Executive Editor, Wiley Internet Technology Letters (2017–2023); Special Issue Editor, ACM TOIT (2023). Co-organizer, Dagstuhl Seminars 21072 and 23081. Two CHI 2025 Honorable Mentions (adaptive industrial robots; responsible ubiquitous personalization, DIS 2025). Finalist, Best Teaching Award, USG (2024).

Narrative. The achievements document three lines of work that ISLANDS brings together. In Web architecture, the PI's work on the Web of Things contributed to a W3C standard that is deployed in industrial

products (10), and his work on hypermedia multi-agent systems introduced signifiers as a run-time abstraction through which agents discover what they can do (6, 9); the agent layer of ISLANDS and its route to standardization build on both. In law and technology, the PI has worked with legal scholars on regulation expressed as technical architecture and on how automation changes the law (4, 5, 8), and his group has audited what platforms disclose under the Digital Services Act (2); this work grounds Islands L and M and the stakeholder analysis. In decentralized data, the PI's group built the secure multi-party computation system over Solid pods on which the confidential references of ISLANDS build (3), and its recent study of how platforms undermine the Web's links states the problem that ISLANDS addresses (1). User-side re-engineering of Web experiences (7) showed early that communities can impose their own guarantees on top of platform defaults.

3. Additional information

Career path. After his doctorate, the PI spent three years in industrial research at Siemens Corporate Technology in Berkeley (2014–2017), where the Web of Things work moved into products. From 2022 to February 2026, he held academic management roles at USG, first as Academic Director and then as Dean of the School of Computer Science, which he built to 10 research groups, more than 120 employees, and about 300 students. He stepped down as Dean in 2026 to refocus on research. The PI has three children (born 2015, 2017, and 2019).

Supervision. Five postdocs mentored into faculty or senior positions (Gómez, Associate Professor, TU Braunschweig; Garcia, Tenure-track Assistant Professor, IT:U Linz; Ciortea, Assistant Professor, USG; Guitton, Swiss Re; Bektaş, Senior Researcher, KSSG). Three PhD graduates in 2024/25 (Ramanathan, Hostettler, Jha), one completing (Vachtsevanou), and six current, including Bekavac and Grau.

Funding. PI or co-PI of 4 SNSF, 2 Innosuisse, 3 EU, and more than 12 industry or foundation projects (3.9 MCHF allocated to the PI's group out of 13.6 MCHF in total), including SNSF HyperAgents (2019–2024), EU IntellIoT (2020–2024), Chips JU AFarEdge (2026–2029, lead of the architecture definition), Innosuisse Flagship WISER (2022–2026), SNSF CoCoDa (2025–2029), SNSF/Innosuisse *Access to Justice* (2026–2030), and TA-SWISS *Sprachmodelle und Kompetenzverschiebungen* (2025–2026).

Contributions to the research community. Active in W3C and IETF standardization; founding member of the W3C Autonomous Agents on the Web Community Group. Steering Committee, International Conference on the Internet of Things (since 2017). Co-organizer of the International Workshop on Hypermedia Multi-Agent Systems (2019–2020) and of the Web of Things workshop series (2012–2017). U.S. Patent US11331803B2 (mixed-reality robot programming, with Siemens). Co-founder of the Future Society Hub (FuSo) at USG, which connects computer science and law.

Application note: Island L and Open Terms Archive

Open Terms Archive (opentermsarchive.org) records every version of the terms of service and privacy policies of digital services "to enable democratic oversight", and the European Commission's DSA enforcement team uses it. Contributors declare, per document, a URL and CSS selectors; robots fetch each document several times a day, store the source as a snapshot, filter it into a textual version, and commit dated versions to public Git repositories, which collections publish as datasets and through a federated API. Open Terms Archive also informs the European Commission about changes in the terms it tracks. Terms published in Island L would make most of this machinery unnecessary. A provider would publish each version of its terms as an immutable, signed record and each change as a superseding record, so that polling robots, change detection, and the dated commits that a third party makes on the provider's behalf would no longer be needed. The Commission would no longer depend on a third party to learn of a change either: it could subscribe to the provider's terms through *Live/Subscription Links* and be notified as soon as the provider publishes a superseding version, which the provider has signed and cannot later deny. Declarations with selectors and filters, and their maintenance whenever a provider redesigns its pages, would also become unnecessary, since the island holds the text itself and not a rendered page. Authenticity would rest on the provider's signature, which a regulator or court can check, instead of on trust in the archivist's crawl. Span addressing would add what the archive lacks today: a verifiable reference to the exact clause of the exact version that a user accepted, which a consumer, regulator, or court can cite. Open Terms Archive would remain necessary for providers that do not join the island, and those that benefit most from opaque terms are the least likely to join. Its collections, analyses, and federated search would remain valuable in any case, as would its memos that explain to the Commission what a change means, but they would work on verified records instead of crawled copies.

Example: a silent correction of a DSA transparency report. An Open Terms Archive investigation by Matti Schneider shows the difference in practice. On 28 August 2026, six months after publication, TikTok repointed the download link of its DSA transparency report for July to December 2025 to a corrected archive, without a public announcement. For terrorist content notices (Category 10d), the number of notices received fell from

77,729 to 43,119 and the median time to take action from 6.82 to 2.43, in a unit the report does not state; the earlier figures had been computed from reports of consumer-related offences. The parent category total was not recalculated, so the corrected file no longer adds up. Open Terms Archive noticed the change only because the report was linked from a page it tracked for another collection, and its analysts had to download both archives and compare them by hand. In Island L, the platform would publish the report as an immutable version and the correction as a superseding version under its signature. Each figure would be span-addressable, so a study that had cited the 77,729 notices through a Verified Link would still resolve to exactly that value, and the island's backlink registry would show the study that a superseding version exists. This answers the question that the investigation leaves open under Implementing Regulation (EU) 2024/2835, which requires versions of transparency reports to be marked and kept accessible: whether one can trace which conclusions rest on obsolete data. Island L would neither prevent a platform from publishing wrong or inconsistent figures, nor make a correction visible if the platform publishes outside the island.